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Making business processes intelligent: Complementarities of combining robotic process automation and AI technologies[☆]

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ABSTRACT

This study examines the firms' combined implementation of Robotic Process Automation (RPA) and Artificial Intelligence (AI) and its association with firm performance using two survey datasets covering large firms worldwide. RPA automates repetitive tasks whereas AI supports cognitive tasks. While previous research typically examines these technologies separately, we draw on complementarity theory to hypothesize that their joint implementation is associated with performance outcomes that exceed the independent benefits of each technology. We find that joint implementation is positively associated with additional revenue growth (linked primarily to price increases) but is not associated with additional cost reduction beyond the separate effects of RPA and AI. These findings challenge the prevailing view that combining RPA and AI technologies primarily enhances performance through increased efficiency, suggesting instead that joint implementation is associated with revenue enhancement. We also consider firms' strategic intent for adopting these technologies and find that among firms implementing both RPA and AI, a stronger revenue growth intent is associated with greater revenue-related outcomes. Our findings provide a baseline analysis of automation complementarities in the pre-Generative-AI era, even as advancements in technology continue and raise new questions.

1. Introduction

Organizations increasingly strive to raise performance by automating operations using digital technologies. These technologies augment firms' other resources and, consistent with the resource-based view of the firm, can enhance their competitive advantage (Bharadwaj, 2000; Oduro et al., 2023; Przegalinska et al., 2025). In this paper, we examine two technologies that modern firms use to automate and augment business processes: *Robotic Process Automation* (*RPA*) and *Artificial Intelligence* (*AI*). We evaluate their association with firm financial performance and explore whether their combined implementation is associated with performance patterns consistent with complementarity.

RPA software technologies are designed to automate repetitive, high-volume tasks that follow predefined rules, such as invoicing and accounts payable processing (Ansari et al. 2019; Chakraborti et al., 2020). AI technologies are algorithmically based applications that perform cognitive tasks associated with humans, often incorporating some form of machine learning (Collins et al., 2021; Dwivedi et al., 2021; Anthony et al., 2023). In practice, many firms adopt both technologies, motivated by the expectation that their combined use may yield complementary performance benefits beyond those achievable by either technology alone. Practitioner discourse often describes such combined use as "intelligent automation" or "intelligent process automation" (Detwiler, 2024; Afrin et al., 2024). However, whether RPA and AI function as complements or substitutes remains an open empirical question.

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Most prior empirical research has examined RPA and AI in isolation, without considering complementarities or the conditions under which such complementarities create additional value (Ameye et al., 2023; Parteka & Kordalska, 2023; Lee et al., 2022; Dixon et al., 2021). This limits our understanding of whether combining these technologies has additional value or whether potential benefits are offset by overlapping functionalities, implementation complexity, or integration challenges. In this paper we investigate whether the joint implementation of RPA and AI technologies is associated with firm performance beyond the independent effects of each technology. We focus on two key firm-level performance dimensions: efficiency-related outcomes, measured as cost reduction, and market-facing outcomes, measured as revenue growth.[5] Our empirical analysis draws on two international survey datasets collected prior to the widespread diffusion of Generative AI (GenAI). Thus, we acknowledge that this study is a baseline analysis of RPA–AI co-implementation in the pre-Generative-AI era that provides a reference point for understanding how automation complementarities may evolve as newer generative technologies emerge.

Prior research relates RPA implementation with cost reduction through several mechanisms, including reduced labor requirements, lower error rates, and increased process speed and scalability. It is also linked with revenue growth caused by freeing organizational capacity to scale operations as well as improved quality, which may support differentiation that allows for increased price or reduced price elasticity of demand. In short, RPA may increase performance by reducing costs and/or increasing revenues. Previous studies of robotization at the firm-level (such as Koch et al., 2021; Ballestar et al., 2020; Acemoglu et al., 2020) document significant performance gains from RPA.

While RPA automates repetitive standard tasks, AI technologies can automate or augment employee work involving human judgement. Prior research relates AI implementation to performance improvements through mechanisms such as improved forecasting, enhanced decision support, and personalization of products and services (Mikalef & Gupta, 2021; Huang & Rust, 2022; Enholm et al., 2022). By overcoming cognitive limits of human decision-making (Jarrahi, 2018), AI can enhance performance through cost reduction (e.g., improved forecasting) and/or revenue growth (e.g., personalization that supports premium pricing) (Anthony et al., 2023; Enholm et al., 2022; Lee et al., 2022; Mikalef & Gupta, 2021).

In sum, while RPA *automates* repetitive tasks by replacing humans with software robots, AI *augments* human decision makers in cognitively more complex non-routine tasks (Raisch & Krakowski, 2021). The benefits of RPA and AI, such as reduced processing time, may be similar, suggesting that they may substitute for one another and affect performance in an additive fashion. Despite potential overlaps, many firms adopt both RPA and AI technologies together, expecting their combination to deliver complementary benefits beyond what either technology can achieve alone. Such joint implementation can enable more advanced automation by combining RPA's strengths in structured, rule-based processes with AI's ability to handle unstructured, cognitive tasks (Afrin et al., 2024; Chakraborti et al., 2020). In practice, the joint use of RPA and AI powers various applications such as virtual assistants, chatbots, compliance monitoring and fraud detection systems, automated hiring and onboarding, intelligent accounting and information processing, and autonomous vehicles (Afrin et al., 2024; Anagnoste, 2018; Bridgelall & Stubbing, 2021; Chakraborti et al., 2020; Detwiler, 2024; Reddy et al., 2019; Roppelt et al., 2025).

Despite practical examples of synergies between RPA and AI, there is little systematic empirical evidence on whether the firms that implement both technologies exhibit greater performance effects and whether these effects materialize as cost reduction, revenue growth, or both. Accordingly, we address the research question: are RPA and AI complementary technologies? We hypothesize that combined implementation of RPA and AI is related to incremental cost reduction and revenue growth compared with the use of either technology alone.[6] Using two international survey datasets, we examine whether firms that have implemented both technologies report incremental performance benefits beyond those associated with each technology individually.

We further consider whether firms' strategic intent behind technology implementation matters. In particular, we examine whether, among firms that implement both RPA and AI, a stronger cost reduction intent is associated with greater cost reduction, and whether a stronger revenue growth intent is associated with greater revenue growth. Based on Lee et al. (2022), we expect that firms with alignment between the intended purpose of technology implementation and the targeted performance outcome are more likely to make informed and targeted investments, thereby reporting greater performance benefits.

Our research contributes novel evidence on the performance implications of business process automation. First, while prior studies have theorized possible interactions between AI and RPA, we present empirical evidence on their combined implementation, leveraging two unique and comprehensive international datasets on large firms' technology investments. Study 1 is based on a 2018 survey of 1219 firms worldwide, capturing *realized* performance outcomes, while Study 2 uses a 2020 survey of 1053 firms, capturing *expected* performance outcomes. These survey data allow us to examine whether joint implementation of RPA and AI is associated with performance outcomes across different samples and time periods, establishing the robustness of our results. Across both datasets, we find consistent evidence that firms using both RPA and AI report higher realized and expected revenue growth compared to firms using only one technology. However, we find no additional cost reduction from combined implementation, beyond the independent benefits of RPA and AI.

Next, we turn to investigating the sources of revenue growth associated with the combined implementation of RPA and AI. Specifically, in Study 1, we decompose revenue change into price effects (e.g., due to price premiums following quality improvement) and volume effects (e.g., due to scaling up) of technology implementation. We find that revenue growth effects are primarily associated with price-based effects, suggesting that combined implementation of RPA and AI may help differentiate the product or service in the eyes of consumers and enhance firms' ability to charge price premiums. In sum, while each technology is associated with cost reduction, their combined implementation is not associated with additional cost reduction. In contrast, combined implementation is associated with incremental revenue growth, supported by price premiums.

Finally, we contribute to the literature by exploring how firms' strategic intent in technology implementation is associated with performance outcomes. We find that among firms using both technologies, those with a stronger revenue growth intent exhibit stronger revenue growth, whereas no comparable pattern emerges for cost reduction. Together, these findings suggest that the performance implications of joint RPA–AI implementation depend critically on the strategic intent guiding technology implementation.

[5] Following prior research on digital technologies and firm performance (e.g., Mukhopadhyay et al., 1995; Melville et al., 2004; Barua et al., 2004), we measure performance along two key dimensions, cost reduction and revenue growth, which are outcomes most directly associated with technology implementation. Cost reduction reflects internal operational gains, while revenue growth captures the combined effect of internal capacity improvements and external market factors, such as increased sales volume and pricing power.

[6] In our study, complementarities are theorized at the firm level to derive from the combined presence of RPA and AI, consistent with complementarity theory (Brynjolfsson & Milgrom, 2013). However, empirically we cannot observe whether these technologies interact within the same processes; if they are implemented in entirely separate parts of the organization, the potential for synergy may not be realized and we would not observe any complementary effects of these technologies.

2. Hypothesis development

2.1. RPA and firm performance

With the rapid development of new technologies, firms increasingly automate their business processes to enhance performance. *Robotic Process Automation* (*RPA*) is one of the foundational automation technologies, designed to automate standard, high-volume repetitive business processes using predefined rules (Ansari et al., 2019; Chakraborti et al., 2020). RPA software can execute existing tasks in systems such as billing and accounts payable, which involve rule-driven, routine activities requiring little human judgment (Aguirre & Rodriguez, 2017). RPA reduces human error, increases speed, and minimizes manual intervention, which translates into efficiency-related improvements through reduction of labor and error-related costs (Anagnoste, 2017; Ansari et al. 2019).

Importantly, the benefits of RPA are not limited to cost reduction. RPA may also be associated with revenue-related outcomes through enhancing scalability. By automating repetitive tasks and removing process bottlenecks, RPA can free up organizational capacity without proportionally increasing costs (Anagnoste, 2017). This improved scalability may allow firms to serve more customers and expand activity levels, thereby supporting revenue growth. For example, in sales processes, RPA can automate routine tasks such as email follow-ups, lead nurturing and quote generation, enabling the sales team to focus on higher-value activities that contribute to closing more deals faster (González, 2023). Taken together, prior literature suggests that RPA contributes to firm performance both by reducing cost and by enabling revenue growth.

2.2. AI and firm performance

Artificial Intelligence (AI) technologies differ from RPA in that they enable machines to perform cognitive tasks traditionally requiring human intelligence, such as reasoning, learning from experience, recognizing patterns, and adapting to new situations (Collins et al., 2021; Dwivedi et al., 2021; Anthony et al., 2023). According to Russell and Norvig (2021), p. vii), AI is "the study of (intelligent) agents that receive percepts from the environment and perform actions," highlighting its capacity to process information and make decisions. AI encompasses technologies such as machine learning, natural language processing, and predictive analytics. Unlike rule-based automation in RPA, AI can learn from data and adapt over time autonomously. That is, AI systems can self-learn, allowing them to operate in more complex and unstructured environments (Huang & Rust, 2018). Specifically, AI *augments* human decision makers in cognitively more complex non-routine tasks (Raisch & Krakowski, 2021). This augmentation — the integration of AI with human expertise — can enhance decisions and optimize processes (Schmidt et al., 2020; Enholm et al., 2022). As a result, AI can offer benefits of both efficiency improvement and revenue growth.

Prior research suggests that AI can improve efficiency-related outcomes that drive cost reduction. By improving decision quality and enabling data-driven optimization of workflows, and smarter resource allocation, AI can reduce operational inefficiencies (González, 2023). For example, in inventory management, AI can enhance demand forecasting by analyzing large amounts of data such as historical sales, inventory records, and market trends, and by identifying patterns that are difficult for humans to detect. This proactive strategy enables businesses to plan efficiently, ensuring they have the right inventory levels to meet customer needs while avoiding surplus stock that results in higher storage costs and potential waste (Praveen et al., 2020). Additionally, AI can reduce operational costs by augmenting humans in simple decision-making tasks and by providing input that improves the efficiency of more complex and creative tasks (Mikalef & Gupta, 2021; Huang & Rust, 2022). Through these mechanisms, AI contributes to improved planning accuracy and reduced resource waste, which translate into lower operational costs.

At the same time, AI is also related to revenue growth by enabling firms to offer more differentiated and personalized offerings (Duan et al., 2019). For instance, the fashion clothing company Gap uses predictive analytics to identify fashion trends, enabling human designers to design clothing that better matches customer preferences. Moreover, AI-enhanced decision making contributes to revenue growth through AI-powered product recommendations, dynamic pricing, and personalized marketing campaigns that can enhance customer experiences and increase willingness to pay (Huang & Rust, 2022; Mikalef & Gupta, 2021). In addition, AI can support innovation and new business models, further contributing to revenue growth (Duan et al., 2019). Overall, prior literature suggests that AI contributes to firm performance through both cost reduction and revenue-enhancing mechanisms.

In recent years, a new class of AI technologies, *Generative AI* (*GenAI*) has emerged, characterized by its ability not only to analyze data but also to generate new content, ideas, and solutions. Advanced large language models such as ChatGPT exemplify this shift from traditional AI, which focuses on prediction and classification, to generative AI, which creates novel outputs (Roberts & Candi, 2024). Firms increasingly view GenAI as an emerging technology distinct from traditional AI, with particular relevance for radical innovation and knowledge-intensive work. However, the diffusion of GenAI is still in its early stages, and most large-scale organizational automation initiatives to date have relied primarily on earlier forms of AI (Roberts & Candi, 2024; Hughes et al., 2026). Our empirical data precede the emergence of GenAI, and our analysis, therefore, captures how cost and revenue performance are associated with traditional AI technologies that dominated business process automation prior to the GenAI inflection point. This provides a foundation for future research on how generative technologies may further transform automation in organizations.

2.3. Combined implementation of RPA and AI

While RPA and AI have been studied largely in isolation, firms increasingly adopt both technologies together. RPA and AI differ in their core capabilities: RPA excels at automating structured, rule-based tasks, while AI augments human judgment in unstructured, data-rich environments (Ansari et al. 2019; Chakraborti et al., 2020). Because these capabilities are distinct, they may be complementary.

According to complementarity theory (Brynjolfsson & Milgrom, 2013), combining resources that provide distinct and mutually reinforcing functions can create additional value beyond their independent effects. RPA and AI exemplify this. First, engagement with RPA may set the stage or create the conditions necessary for applying AI more effectively such that as routine tasks become automated, more complex tasks can be augmented by AI. Second, automation by RPA in one task can spill over, enabling augmentation of adjacent tasks by AI. For example, when RPA is implemented to automate order taking in restaurants, the adjacent task of providing meal recommendations can be enabled through AI-recommendation systems (Raisch & Krakowski, 2021). In practice, the combined use of RPA with AI, often described as "intelligent process automation", refers to automation architectures that integrate RPA's rule-based execution with AI's cognitive decision-making capabilities (Anagnoste, 2018; Reddy et al., 2019; Chakraborti et al., 2020). In this context, combining RPA and AI can allow firms to unlock new opportunities for cost reduction and revenue growth by integrating the repetitive task expertise of RPA software bots with the cognitive skills of AI. Thus, consistent with complementarity theory (Brynjolfsson & Milgrom, 2013), the performance impact of each technology is amplified when combined with the other.

The joint implementation of RPA with AI has been observed in practice in personal assistants like Alexa, Cortana, Google Assistant, fraud detection systems, and autonomous vehicles (Anagnoste, 2018; Reddy et al., 2019; Bridgelall & Stubbing, 2021). Firm-specific applications can be seen in several areas (Detwiler, 2024) such as (1) customer service, where chatbots in online training, customer service, and cognitive therapy benefit from having a human-like touch to go beyond machine-like interactions (Go & Sundar, 2019; Huang & Rust, 2022); (2) human resources, where the combination of technologies can streamline hiring processes with AI performing background checks on newly hired employees and bots helping onboard these employees by provisioning accounts, populating databases, and preparing orientation materials tailored to each worker; (3) accounting and reporting, where AI can extract key details from supplier or customer invoices, such as amounts owed, due dates, and purchase order numbers. In accounts payable, RPA software bots can use this data to verify orders, calculate totals, submit payments for approval, and process approved payments. For accounts receivable, RPA can automate payment reminders, reconcile payments, and flag late payments for collection. The combination of technologies can also help ensure compliance with financial regulations, such that AI can detect if the same employee checks and approves an invoice, flagging it as an anomaly and deploying an RPA bot to prevent payment.

Based on the complementary features of RPA and AI, we argue that the joint implementation of RPA and AI may be associated with additional cost reduction beyond the independent effects of each technology. RPA can automate repetitive tasks and standardize processes, while AI can identify inefficiencies, optimize workflows, and support more complex decision-making (Anagnoste, 2017; Ansari et al. 2019; Mikalef & Gupta, 2021; Huang & Rust, 2022). Together, these capabilities may enable firms to further reduce processing time, improve accuracy, and allocate resources more efficiently. Thus, we hypothesize that joint implementation is associated with enhanced efficiency improvements and resulting cost reductions compared to using either technology alone.

H1. A combined implementation of RPA and AI is positively associated with cost reduction.

From a revenue perspective, combined implementation of RPA and AI may also support additional revenue growth through two underlying mechanisms: price change and volume change. First, joint implementation may enhance firms' ability to generate revenue through price premiums. AI enables firms to improve product and service differentiation by leveraging data to personalize offerings, optimize pricing, and enhance decision-making (Huang & Rust, 2022; Mikalef & Gupta, 2021). At the same time, RPA contributes by ensuring consistent, reliable, and timely execution of customer-facing processes, reducing service errors and improving responsiveness. By combining AI-driven differentiation with RPA-enabled process reliability and service quality, firms can enhance the customer value proposition. This combination can increase customers' willingness to pay, enabling firms to charge higher prices for differentiated and more reliable offerings.

Second, joint implementation of RPA and AI may support revenue growth through increased activity and sales volume. RPA can automate repetitive tasks and remove operational bottlenecks, while AI can improve decision-making and process optimization in more complex tasks. Together, next to an enhanced customer value proposition that may spur demand, these technologies can expand operational capacity and improve scalability, allowing firms to handle greater activity levels in the form of more customers or transactions. As an example, in the financial industry, intelligent automation of KYC (know your customer) processes can help overcome human capacity constraints associated with manual verification, enabling firms to onboard more customers more quickly (enhancing service quality perceptions) and increase transaction volumes, thereby supporting revenue growth.

Together, these two mechanisms suggest that joint implementation of RPA and AI may be associated with greater revenue growth relative to independent implementation. We, therefore, hypothesize:

H2. A combined implementation of RPA and AI is positively associated with revenue growth.

2.4. Strategic intent and performance outcomes from combined implementation

While firms may jointly implement RPA and AI, they may differ in the extent to which they realize performance benefits from these technologies in the form of cost reduction (H1) and revenue growth (H2). In particular, complementarity between RPA and AI does not arise automatically; its realization depends on how and why these technologies are implemented.

In practice, RPA and AI may be implemented in separate business units or operational processes, limiting opportunities for the technologies to interact. In such cases, the two technologies may be associated with performance independently, but not jointly, especially when they are spatially and/or temporally separated across operational processes. Moreover, the technologies may overlap in realizing the same objective, such as reducing processing times and improving service speed, and thus substitute rather than complement each other in certain contexts. In other cases, firms might even apply AI to routine tasks better suited for RPA, again potentially substituting for each other rather than complementing each other. Such overlap might arise due to poor process design, lack of strategic alignment, or immature implementation decisions, and can limit the realization of complementarities.

Even when complementarities are theoretically possible, the organizational and technical complexity and costs of jointly implementing RPA and AI may inhibit firms from unlocking them (Afrin et al., 2024). Because both technologies are complex to deploy even individually, their joint implementation might be more complex, requiring substantial expertise, coordination, and investment. Consequently, it may involve prohibitive costs, which, with poor implementation, may outweigh realized performance gains, such as when organizations lack a proper understanding of complex technologies, underestimate the associated challenges, or implement them without a proper strategy (Fountaine et al., 2019).

According to resource orchestration theory (Oduro et al., 2023), resources contribute to performance not merely through their presence, but through purposeful deployment aligned with strategic objectives. Thus, the way in which technologies are jointly employed is as important as the technologies adopted (Oduro et al., 2023). Achieving performance benefits from resources depends not only on their presence but on how they are purposefully deployed and aligned with strategic goals. We, therefore, consider that performance gains from combined implementation of RPA and AI are more likely to materialize when firms adopt these technologies based on a clear, aligned digitization strategy.

Prior research suggests that technologies generate higher returns when they are adopted with clear intent and are integrated into organizational processes aligned with desired outcomes (Brynjolfsson et al., 2019; Lee et al., 2022). For instance, firms that pursue a focused, exclusive R&D strategy that is tailored to their goals realize greater performance gains from AI technologies, such as revenue growth (Lee et al., 2022). Successful implementation of any technology requires organizational adaptations or changes before firms can transform their businesses to fully realize benefits (Fountaine et al., 2019). This involves investing in both tangible and intangible assets, comprising human capital investment and the co-invention of new business models and products (Ransbotham et al., 2020; Brynjolfsson et al., 2019). Firms with a clear strategy for technology implementation can make informed, targeted investments for this purpose (Lee et al., 2022).

Building on this perspective, we expect variation in strategic intent among firms with joint implementation of RPA and AI to be associated with differences in performance outcomes. Even when firms adopt both technologies, the importance assigned ex ante to specific objectives, such as cost reduction or revenue growth, shapes how these technologies are deployed, integrated, and utilized within the organization. We expect that firms that implement these technologies with a clear purpose and strategy are more likely to deploy and invest in resources, such as digitalization expertise, that will enable them to overcome the organizational and technical challenges of integration. Such clarity allows firms to (re-)direct resources effectively to accomplish specific objectives of implementing technology.

When firms emphasize cost reduction for technology adoption, we expect that they will be more likely to deploy RPA and AI in ways that support efficiency improvements. Such firms may prioritize process standardization, automation of high-cost activities, and integration of AI-driven decision-making into operational workflows. By aligning implementation efforts with cost reduction objectives, these firms are more likely to direct resources toward reducing operational costs and realize better cost reduction outcomes. In contrast, firms with a weaker cost reduction intent may deploy the same technologies without systematically targeting efficiency improvements, limiting realized cost reductions. Accordingly, we hypothesize that among firms with a joint implementation of RPA and AI, variation in cost reduction intent will be associated with differences in cost reduction outcomes.

H3. Among firms with joint implementation of RPA and AI, a greater cost reduction intent is positively related to cost reduction.

In a similar fashion to H3, we expect that firms that emphasize revenue growth in their technology adoption are more likely to deploy RPA and AI in ways that support revenue generation. Such firms may target AI use toward enhancing customer value through personalization, pricing, and product or service differentiation, while leveraging the combination with RPA to ensure efficient, scalable, and reliable execution of these activities. This alignment can support both sales price and volume by enabling firms to deliver differentiated offerings and expand operational capacity. In contrast, we expect that firms with a weaker revenue growth intent may not systematically deploy these technologies toward customer-facing or market-expanding activities, resulting in more limited revenue growth. Accordingly, we expect that among firms with joint implementation of RPA and AI, variation in revenue growth intent will be associated with differences in revenue growth.

H4. Among firms with joint implementation of RPA and AI, a greater revenue growth intent is positively related to revenue growth.

3. Overview of empirical methods

To test our hypotheses, we use two international survey datasets collected from large firms in 2018 and 2020 (referred to as Study 1 and Study 2, respectively). Data collection for the two surveys was commissioned by Deloitte Consulting LLP (hereafter, "Deloitte") from a large, reputed international third-party survey research firm. Due to differing survey objectives, the geographical scope varies: the 2018 survey covers large firms in 24 countries, and the 2020 survey covers large firms in 14 (partially overlapping) countries. The contracted survey research firm was tasked with obtaining a minimum number of qualified responses per targeted geographic region. A qualified response is one in which a participant passes all screening questions (e.g., relevant expertise and experience) and substantially completes the survey. The academic authors were invited to provide input to both surveys and several, but not all suggestions were adopted. The surveys were administered in the (common business) language of the respondent's country after appropriate testing of the stability of the instrument across versions (e.g., translation followed by back-translation to ensure that meaning was not altered). For both surveys, invited respondents were not informed that Deloitte was the sponsor.

Both surveys include relevant measures that enable us to test H1 and H2 on the cost reduction and revenue growth outcomes of combined implementation of RPA and AI. Study 1 also uniquely enables us to test H3 and H4, concerning the strategic intent for technology implementation. Since the two studies differ in scope, measures, and timing, they serve as two independent tests of the hypotheses.

We estimate the following empirical models to test hypotheses 1 and 2:

$$Y_i = c + \beta_1 RPA_i + \beta_2 AI_i + \lambda^T X_i + \sum_{j=1}^{J-1} \gamma_j Industry_{ij} + \sum_{k=1}^{K-1} \delta_k Country_{ik} + \varepsilon_i \quad (1)$$

$$Y_i = c + \beta_1 RPA_i + \beta_2 AI_i + \beta_3 RPA\_AI\_Joint_i + \lambda^T X_i + \sum_{j=1}^{J-1} \gamma_j Industry_{ij} + \sum_{k=1}^{K-1} \delta_k Country_{ik} + \varepsilon_i \quad (2)$$

Model (1) estimates the independent effects of RPA and AI implementation. Model (2) introduces an interaction term between RPA and AI implementation ($RPA \times AI$), denoted as *RPA_AI_Joint,* which captures whether the joint implementation of both technologies is associated with performance effects beyond their independent effects. The dependent variable Y_i represents performance outcome of interest and is measured first as cost reduction and then as revenue growth in firm *i*. The two survey datasets allow us to use alternative ways to measure Y_i. In Study 1, we measure Y_i as realized annual cost reduction (*Cost reduction*) and realized revenue growth in the past 24 months (*Revenue change*) (see Table 2 for variable descriptions of Study 1). In Study 2, we measure Y_i as expected cost reduction (*Expected cost reduction*) and expected revenue growth (*Expected revenue growth*) (see Table 10 for variable descriptions of Study 2). Although the measures differ, this design allows us to assess whether realized gains associated with the combined implementation of RPA and AI (Study 1) converge with the gains that managers expect from it (Study 2).

In both models, c is the intercept term. β_1 and β_2 are the coefficients for RPA and AI technology implementation in firm *i*, respectively. In model (2), β_3 is the coefficient on the interaction term (*RPA_AI_Joint*) and represents the association between the joint implementation of RPA and AI with the performance outcome of interest, Y_i. X_i is the vector of firm-level control variables for firm *i* with associated coefficients λ^T. $Industry_{ij}$ is an industry indicator (1 if firm *i* is in industry *j*, 0 otherwise) with coefficients γ_j. $Country_{ik}$ is a country indicator (1 if firm *i* is in country *k*, 0 otherwise) with coefficients δ_k. ε_i is the error term capturing unobserved factors associated with Y_i.

We test hypotheses 3 and 4 by examining whether variation in firms' strategic intent for technology adoption within the subsample of firms using both technologies is associated with the performance outcomes of interest. Strategic intent is operationalized using firms' reported importance of different reasons for adopting each technology, as described in the variable measurement section.

Consistent with H3 and H4, our empirical approach focuses on the subsample of firms that have implemented RPA, AI, or both technologies. Within each subsample, we examine whether differences in the importance assigned to cost reduction and revenue growth intents are associated with corresponding differences in performance outcomes. This approach allows us to isolate variation in strategic intent among firms using these technologies, rather than comparing across adopters and non-adopters. We estimate the following model:

Table 1
Sample descriptive statistics.

Panel A: Firm size, industry, and governance (N = 1219)					
Revenues in millions (€)[1]	*# obs*	*% obs*	*# Employees*	*# obs*	*% obs*
175 – 250	101	8.29	1–1000	97	7.96
250 – 400	77	6.32	1000 – 2500	185	15.18
400 – 600	90	7.38	2500 – 5000	180	14.77
600 – 1000	103	8.45	5000 – 10,000	228	18.70
1000 – 3000	186	15.26	10,000 – 25,000	163	13.37
3000 – 4000	107	8.78	25,000 – 50,000	123	10.09
4000 – 8000	158	12.96	50,000 – 100,000	134	10.99
8000 – 15,000	135	11.07	100,000 and more	109	8.94
15,000 – 25,000	95	7.79			
25,000 – 50,000	64	5.25			
Over 50,000	103	8.45			
Principal industry					
Consumer goods	159	13.04	Life sciences & health care	157	12.88
Industrial	163	13.37	Public sector	87	7.14
Energy & resources	144	11.81	Technology, media, communication	221	18.13
Financial services	279	22.89	Other	9	0.73
Public firm					
Yes	273	22.40			
No	946	77.60			

Panel B: Firm country of origin (N = 1219)					
Country	*# obs*	*% of sample*	*Country*	*# obs*	*% of sample*
Australia	40	3.28	Italy	41	3.36
Belgium	15	1.23	Japan	40	3.28
Brazil	70	5.74	Mexico	72	5.91
Canada	50	4.10	Netherlands	40	3.28
Chile	25	2.05	New Zealand	30	2.46
China	80	6.56	Norway	13	1.07
Denmark	10	0.82	Singapore	31	2.54
Finland	15	1.23	South Africa	30	2.46
France	50	4.10	Spain	50	4.10
Germany	70	5.74	Sweden	10	0.82
Hong Kong	30	2.46	UK	100	8.20
India	81	6.64	USA	226	18.54

[1]U.S. firms with less than $1.5 billion and firms in other countries with less than €175 million (after currency conversion) in revenues are screened from completing the survey. To combine surveys that use other currencies we convert to Euro using an average conversion rate at the time of the survey.

$$Y_i = c + \beta_1 Intent\_RPA_i + \beta_2 Intent\_AI_i + \beta_3 Intent\_Joint_i + \lambda^T X_i + \sum_{j=1}^{J-1} \gamma_j Industry_{ij} + \sum_{k=1}^{K-1} \delta_k Country_{ik} + \varepsilon_i \tag{3}$$

where Y_i denotes the performance outcome of interest. To test H3, Y_i represents cost reduction, and the intent variables correspond to cost reduction intent for RPA and AI. To test H4, Y_i represents revenue growth (and, in additional analyses, its components: price change and volume change), and the intent variables correspond to revenue growth intent.

Model (3) is estimated separately across subsamples of firms that have implemented RPA, AI, and both technologies, respectively. In the RPA subsample, the RPA-specific intent variable captures the importance assigned to cost reduction or revenue growth for implementing RPA. Similarly, in the AI subsample, the AI-specific intent variable captures the corresponding strategic intent for implementing AI. In the joint implementation subsample, the joint intent variable, defined as the product of the intent measures for RPA and AI implementation, captures the combined emphasis placed on cost reduction or revenue growth objectives across both technologies. Among firms that have jointly implemented RPA and AI, this approach allows us to assess whether alignment in strategic intent (reflected in a stronger emphasis on cost reduction or revenue growth) is associated with greater corresponding performance benefits, consistent with H3 and H4.

4. Study 1

4.1. Methods

4.1.1. Research design and data collection

The data for Study 1 are from large firms in 24 countries and enable testing all four hypotheses. While for its purposes, Deloitte designed the survey with a focus on firms' practices for cost management, the survey includes questions about firms' technology use and purposes. The contracted survey research firm maintains a comprehensive database of millions of international consumers and business professionals. It connects organizations that aim to set out a survey with qualified and verified target respondents and manages the process of data collection.[7] In developing a respondent panel for a specific survey, the firm maximizes fit with the target respondent group while avoiding over-sampling of respondents to avoid survey exhaustion. After accepting the invitation

[7] Respondents typically opt-in to the survey firm's participant database as part of an online transaction with an unrelated merchant. At the conclusion of a transaction, the merchant offers customers an opportunity to earn some form of reward (e.g., points in a marketing loyalty reward program, discounts for future transactions) in return for completing a survey. After agreeing to complete a survey, customers may be asked whether they wish to receive future invitations to complete surveys in return for rewards. When agreeing to this, they complete a basic demographic survey and are added to the database of the survey firm, with a profile update taking place each time when completing a survey. The survey firm also contacts respondents regularly to verify and update demographic data that are critical to building targeted survey invitations.

Table 2
Variable measurement and summary statistics.

Variable	Description	Obs.	Mean	S.D.	Min	Max
Cost reduction	Total percentage annual cost savings calculated by formula of *Annual cost reduction target percentage * % Target achievement)/ 100*	1193	9.01	9.16	0	50
Revenue change	Percentage change in revenue over the past 24 months	1219	8.64	14.25	−50	50
RPA	Dummy variable is 1 when RPA is implemented, otherwise 0	1219	0.24	0.43	0	1
AI	Dummy variable is 1 when AI or Cognitive technology is implemented, otherwise 0	1219	0.25	0.43	0	1
Price change	Categorical measure of the annual change in output prices across the firm's product lines over the past 24 months	1218	4.46	6.08	−20	20
Volume change	Calculated by formula: *100* [(1 + percent revenue change) / (1 + price change) − 1]*	1218	3.90	10.97	−58.33	59.57
Cost_RPA	The average of the importance of two cost reduction intents (i.e., reduce labor and all other costs) for using RPA	292	4.38	0.74	1	5
Cost_AI	The average of importance of two cost reduction intents (i.e., reduce labor and all other costs) for using AI	301	4.29	0.80	1	5
Rev_RPA	The average of the importance of two revenue intents (i.e., increase revenue and enhance product/service capabilities) for using RPA	292	4.39	0.75	1	5
Rev_AI	The average of the importance of two revenue intents (i.e., increase revenue and enhance product/service capabilities) for using AI	301	4.44	0.65	1.5	5
Digital leader	Dummy variable is 1 when there is digital strategy leader, otherwise 0	1219	0.80	0.40	0	1
CEO_President	Dummy variable is 1 when respondent is CEO or President of the company, otherwise 0	1219	0.24	0.43	0	1
CFO_COO	Dummy variable is 1 when respondent is CFO or COO of the company, otherwise 0	1219	0.21	0.41	0	1
New CEO	Dummy variable is 1 when there is new CEO, otherwise 0	1219	0.53	0.50	0	1
Acquisition	Dummy variable is 1 when there is an acquisition, otherwise 0	1219	0.53	0.50	0	1
Sales dispersion	Herfindahl index, calculated as the sum of the squared percentages of regional revenues subtracted from one	1219	0.57	0.32	0	0.87
Total revenue	Log of total revenues	1219	21.93	1.69	19.17	24.64
Employees	Log of employees	1219	8.27	2.81	0	11.51
Public firm	Dummy variable is 1 for firms that are publicly traded, otherwise 0	1219	0.22	0.42	0	1

to participate, respondents first answered screening questions to ensure eligibility (i.e., relevant expertise and experience). Screening questions also restricted participation to U.S. firms with sales revenues of greater than $1.5 billion and to firms in other countries with revenues (after currency conversion) greater than €175 million. These sampling criteria result in representativeness of the data and statistical results relating to large firms, across a wide range of industries and a range of countries. The survey research firm monitored survey completion and sent invitations as needed to meet coverage agreements (e.g., per targeted country). The process ensured that the desired sample size and coverage was realized, although its dynamic nature means we cannot compute a conventional survey response rate.

4.1.2. Survey respondents and sample description

Across all 24 countries, there are 1219 complete valid survey

Table 3
Overview of summary statistics for Cost reduction, Revenue change and Price change.

Panel A: Cost reduction based on annual cost reduction target (%) and realized savings			
Cost reduction target (%)	***Scale value***	***# obs***	***% obs***
< 5%	1%	72	5.9
5–10%	5%	295	24.2
10–20%	10%	450	36.9
20–30%	20%	234	19.2
30–50%	30%	93	7.6
50% or more	50%	49	4.0
No specific target established	-	26	2.1
Savings realized relative to target (i.e., % Target achievement)			
No savings	0%	16	1.3
Up to a quarter (25%) of savings target	12.5%	113	9.3
Up to half (50%) of savings target	37.5%	190	15.6
Up to three quarters (75%) of savings target	62.5%	333	27.3
Between 75% and 100% of target	87.5%	333	27.3
Full realization (100%) of savings target or more	100%	234	19.2
		Mean	St.dev.
Cost reduction: (Cost reduction target % * % Target achievement)/100		***9.01***	***9.16***

Panel B: Revenue change realized in the prior 24 months							
	Scale value	***# obs***	***% obs***		***Scale value***	***# obs***	***% obs***
Decreased more than 50%	−50%	7	0.6	Increased more than 50%	50%	70	5.7
Decreased 31–50%	−31%	12	1.0	Increased 31–50%	31%	91	7.5
Decreased 11–30%	−11%	11	0.9	Increased 11–30%	11%	261	21.4
Decreased 6–10%	−6%	29	2.4	Increased 6–10%	6%	353	29.0
Decreased 1–5%	−1%	32	2.6	Increased 1–5%	1%	271	22.2
Remained the same	0%	82	6.7				

Panel C: Price change realized in the prior 24 months							
	Scale value	***# obs***	***% obs***		***Scale value***	***# obs***	***% obs***
Declined more than 20%	−20%	7	0.6	Grew between 0% and 5%	1%	385	31.6
Declined between 10% and 20%	−10%	9	0.7	Grew between 5% and 10%	5%	338	27.8
Declined between 5% and 10%	−5%	27	2.2	Grew between 10% and 20%	10%	176	14.5
Declined between 0% and 5%	−1%	42	3.5	Grew above 20%	20%	76	6.2
No change	0%	158	13.0				

responses. Table 1, Panel A reports the distribution of sample firms over size categories in revenues (in Euro) and number of employees, across industries, and with respect to governance aspects. The sample is well-dispersed across size categories and industries. Approximately a quarter of the sample firms is publicly listed. Panel B reports firms' country of origin.

4.1.3. *Survey design and variable measurement*

The survey was an adjusted version of a survey completed in 2016 by an (independent) international sample of over 1000 large firms. Similar to a pilot test, this provided thorough insight into the clarity and understandability of recurring questions, and respondents' ability or knowledge to address questions. The survey, including newly added questions, was reviewed thoroughly by the academic authors and by the survey firm. The survey presented questions in a fixed order, beginning with questions about the firm's current conditions and then progressing to questions about the future. Questions within a question block were presented in random order to mitigate ordering effects. Most questions asked for objective and relatively recent information such as the presence or absence of specific technologies and percentages (e.g., sales change, cost reduction). Compared to perceptual measures, this mitigates potential biases such as hindsight bias and common method bias (Golden, 1992). We note that this nature of measurement also precludes providing statistical evidence on construct validity, which is primarily based on face validity. Below we describe the measurement of variables used in the hypothesis tests. Table 2 summarizes the measures and reports descriptive statistics.

Cost reduction is used to measure the firm's realized annual cost reduction using two survey items: (1) the firm's annual cost reduction target as a percentage of costs in the prior fiscal year, and (2) the percent target achievement (Table 3, Panel A). Both items are reported in percentage intervals, which we convert into numerical values using the lower bound of each interval for cost reduction targets and the midpoint of each interval for target achievement.[8] We compute the firm's realized cost reduction in percentage as the product of the cost reduction target percentage and the corresponding percent target achievement. For instance, a 10% cost reduction target with 62.5% achievement represents a realized cost reduction of 6.25%. Variation in both the cost reduction target and target achievement items is high, and so is variation in the measure of realized cost reduction. A small number of firms indicated having no cost reduction target, reducing the sample to 1193 firms for analyses including this variable.

Revenue change is used to capture realized revenue growth, measured as the firm's approximate annual change in revenue over the past 24 months. The survey scale measures changes in revenue in percentage intervals (e.g., "increased 31–50%" or "decreased 31–50%"). We convert these intervals into numerical values using the lower absolute value of each interval and then applying a positive or negative sign to reflect the direction of change. For example, "increased 31–50%" and "decreased 31–50%" are coded as + 31% and –31%, respectively. This approach allows unambiguous coding of the open-ended lowest and highest scale categories (i.e., –50% for "decreased more than 50%" and +50% for "increased more than 50%"). It also ensures a symmetric and conservative coding scheme that avoids overstating the magnitude of

[8] The annual cost reduction target question is framed as an annual cost reduction target for the firm's improvement initiatives as compared with prior year's costs for prior year output, and is captured with six response categories, ranging from less than 5–50% or more. Given the scale construction in which the last interval option (50% or more) has no specified range, we use the lowest bound of each interval (e.g., 5–10% is coded as 5%). This provides a conservative estimate of the cost reduction target (e.g., 1%, 5%, 10%…50%). Observations indicating "no specific target established" are excluded, as the variable measure cannot be constructed for these cases.

Table 4
Summary statistics for technology implementation and intent of technology implementation.

Panel A: Technology implementation in the prior 24 months			
Robotic Process Automation (RPA)	***Artificial Intelligence (AI)***		
	Not implemented[a]	Implemented	
Not implemented[a]	777	140	*917*
Implemented	137	165	*302*
	914	*305*	*1219*

Panel B: Strategic intent of technology implementation						
	RPA			***AI***		
	Obs.	*Mean*	*S.D.*	*Obs.*	*Mean*	*S.D.*
Reduce cost (excluding labor)	292	4.45	0.83	301	4.26	0.91
Reduce labor cost	292	4.30	0.85	301	4.33	0.89
Increase revenue	292	4.39	0.89	301	4.43	0.76
Enhance product/service capabilities	292	4.38	0.81	301	4.45	0.79

[a] Not implemented includes not implemented and not planned; not implemented but planned; or in process of implementation.
Note: Group means are the average importance of each intent for firms using RPA, and AI, respectively.

both positive and negative changes (Table 3, Panel B). As a result, *Revenue change* ranges from −50–50 % where negative values indicate revenue decline and positive values indicate revenue growth. Most firms experienced revenue growth, although 7.5 % report a decline and 6.7 % report no change.

To analyze the mechanisms of revenue change, we decompose *Revenue change* into *Price change* and *Volume change*. *Price change* is measured as the annual change in output prices across the firm's product lines over the past 24 months, and is reported in directional percentage intervals ranging from "20 % or greater reduction" to "20 % or greater increase" (Table 3, Panel C). Consistent with the treatment of revenue change, we convert these intervals using the lower absolute value of each interval, with a positive or negative sign reflecting the direction of change (e.g., +10% for "10–20% increase" and –10% for "10–20% decrease"). For interval categories bounded at zero (e.g., "declined between 0–5%" and "grew between 0–5%"), we assign a value of ±1% rather than 0% to preserve the direction of change while maintaining a conservative coding approach. To obtain a clean measure of the experienced change in volume, we remove the firm-specific output price changes from revenue change, computing *Volume change* as: 100*[(1 + percent revenue change) / (1 + percent price change) – 1]. In this calculation, percent changes are entered in decimal form, so that 10% is coded as 0.10 and −1% as −0.01. Multiplying by 100 expresses *Volume change* in percentage-point units, making it directly comparable in scale to revenue change and price change variables.

It is important to note that our key performance variables (cost reduction, revenue change, and price change) are based on categorical survey ranges rather than continuous measures. Although this provides a pragmatic and factual approach to operationalize performance using the available data, it does introduce some measurement imprecision as the exact values within each interval are unknown. The coding rules adopted reflect the structure of the underlying survey questions. For directional variables such as revenue and price change, we use the lower absolute value of each interval to obtain conservative and symmetric estimates of change. In contrast, percent target achievement is non-directional and represents realized performance within a bounded interval with a maximum of 100%; therefore, we use the midpoint of each interval to approximate the central tendency. As a robustness check, we confirm that alternative coding approaches yield qualitatively similar results.[9]

Robotic Process Automation (*RPA)* is measured using a factual survey item asking whether the firm had implemented RPA. We measure *RPA* with an indicator variable set to 1 if RPA is implemented and 0 if RPA is not implemented and not planned; not implemented but planned; or in process of implementation.[10] Similarly, *Artificial Intelligence (AI)* is an indicator variable set to 1 if AI is implemented and 0 otherwise. As Table 2 shows, 24.77% and 25% of the sample firms had implemented RPA and AI, respectively. The cross-tabulation reported in Table 4, Panel A shows implementation of both RPA and AI technologies by 165 firms (i.e., 13.53% of the sample).

We operationalize joint implementation of RPA and AI as an interaction term *RPA* × *AI*, denoted as *RPA_AI_Joint*. This indicator is coded 1 when both technologies are implemented and 0 otherwise. This operationalization aligns with prior research on IT complementarities that similarly infers synergy potential from firm-level joint implementation of complementary technologies (e.g., Bresnahan et al., 2002; Aral et al., 2012; Brynjolfsson & Hitt, 2000). While the survey does not directly observe process-level integration, joint implementation is a necessary (though not sufficient) precondition for complementary effects to arise. Moreover, our tests of H3 and H4 (showing whether firms with aligned strategic intent for implementing RPA and AI achieve stronger performance outcomes) provide additional evidence that joint implementation often reflects intentional integration rather than unrelated use.

Respondents described their firm's strategic intent for using the technologies through a Likert-type scale of varying importance (1 = unimportant, 5 = very important) for a set of purposes (e.g., reducing labor cost, increasing revenue, enhancing product/service capabilities). Table 4, Panel B presents summary statistics for these measures.

To measure cost reduction intent, we compute *Cost_RPA* as the average importance assigned to two cost-related objectives for firms using RPA: reducing labor cost and reducing other costs ($r = 0.54$, $p < 0.01$). Similarly, *Cost_AI* is constructed as the average importance assigned to the same objectives for firms using AI ($r = 0.60$, $p < 0.01$). The means of *Cost_RPA* and *Cost_AI* are 4.38, and 4.29 respectively (Table 2), indicating that cost reduction is a highly important strategic intent for firms using these technologies. Table 2 also shows there is quite some variation among firms for using these technologies to reduce costs, enabling us to test H3.

To measure revenue growth intent, we compute *Rev_RPA,* and *Rev_AI* as the average importance assigned to two revenue-enhancing objectives for using these technologies: increasing revenue, and enhancing product/service capabilities ($r = 0.56$ and $r = 0.42$, respectively, both $p < 0.01$). The mean values of 4.39, and 4.44 for *Rev_RPA,* and *Rev_AI,* respectively (Table 2), indicate that revenue growth is a highly important strategic intent for firms using each technology, with sufficient variation among firms to examine H4.

To capture the joint emphasis placed on cost reduction intent across both technologies, we construct the interaction term *Cost_Joint* as the product of the mean-centered measures of *Cost_RPA* and *Cost_AI*. This variable reflects the cost reduction intent of firms using both RPA and AI. Similarly, we calculate *Rev_Joint* as the product of the mean-centered measures of *Rev_RPA* and *Rev_AI*, capturing the revenue growth intent of firms using both technologies.

As also reported in Table 2, we control for several firm and industry characteristics that could influence technology use and the cost reduction and revenue gains that we examine as outcomes. *Digital leader*

[9] As a robustness check, we verify that alternative coding approaches of using interval midpoints for the cost reduction target, revenue change and price change items yield similar results for the hypothesis tests.

[10] For firms in the process of implementing RPA, it unclear how much of the technology has already been implemented, and whether it is already in operation. We assume that firms are able to use and derive benefits from technologies only after their implementation.

indicates whether the firm has appointed a designated leader for digital strategy (= 1) or not (= 0). Having such a leader, which holds for 80% of sample firms, may profoundly influence the firm's orientation towards and execution of a digital strategy (Davison et al., 2023). This variable thus controls for potential confounding leadership effects that may determine the quality and impact of technology implementation and use.

CEO_President indicates whether the respondent is the firm's CEO or President, which is the case for 24 % of the sample firms. *CFO_COO* indicates if the respondent is the firm's COO or CFO (21 % of the sample). *New CEO* indicates whether the firm had a new CEO appointed in the last 24 months, which was the case for 53 % of firms. New leaders typically initiate new plans and strategies to reduce costs and enhance revenue, which may also involve digitalization initiatives. This indicator thus also controls for leadership and governance effects that could confound the effects of the technology variables.

We further control for *Acquisition*, which indicates whether the firm acquired another firm or was acquired in the last 24 months. Acquisitions can influence both cost and revenue through integration processes and operating synergies (Devos et al., 2009; Rabier, 2017), as well as through the integration and introduction of technologies (Smeulders et al., 2023). The data lack the reasons for CEO change or whether in an acquisition the firm is the acquirer or the acquired. Yet, firms targeted for acquisition because of inefficiencies often have top managers replaced. We, therefore, also interact *New CEO* and *Acquisition* to distinguish the situation in which the firm is more likely to be the acquired and CEO change more likely reflects replacement of an ineffective leader. In such cases, CEO change is more likely to be associated with weak prior performance and a stronger impetus to pursue cost reduction and revenue growth through restructuring and digital transformation initiatives.

As a final control for governance influences, *Public firm* indicates whether firms are publicly traded (24 %). Governance and scrutiny by analysts influence resource management, and pressures to deliver short-term profits may more strongly motivate cost reduction and revenue growth than in privately held firms.[11]

Firms with greater scope and scale may benefit differently from investments in RPA and AI than other firms (Koch et al., 2021; Lee et al., 2022). Accordingly, we also control for firm size and geographical sales dispersion. We take two measures of firm size, based on revenue (with all currencies converted to Euros) and the number of employees worldwide, which may both affect performance (e.g., a larger employment base may offer more opportunity for productivity improvement (Pinnuck & Lillis, 2007), through efficiency or growth benefits from automation or augmentation). For reasons of firm confidentiality, respondents selected intervals for these measures (e.g., 4 = 5000 to less than 10,000 employees). We transform these intervals using the lower bound of each interval, noting that estimation results are similar when instead using the mid-point (with the lower bound for the final open-ended category). For both *Total revenue* and *Employees*, we take the log of the values because the category range expands with size. To measure *sales dispersion*, we compute a Herfindahl index as the sum of the squared percentages of regional revenues in the prior year subtracted from one. Larger values indicate more dispersed sales across regions worldwide.

We include in the analyses industry fixed effects to control for systematic differences across industries that could confound the estimated relationships.[12] We also include country fixed effects based on the firm's

Table 5
Correlation table of firm-level variables This table reports the Pearson correlation between pairs of firm-level variables. Cells of paired indicator variables (Variables 5, 6, 7, 8, 9, 10, 11, and 15) contain tetrachoric correlations. We exclude from the table the subsample measures: Cost_RPA, Rev_RPA, Cost_AI, and Rev_AI.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14
(1) Cost reduction	1.000													
(2) Revenue change	0.265***	1.000												
(3) Price change	0.344***	0.589***	1.000											
(4) Volume change	0.124***	0.895***	0.174***	1.000										
(5) RPA	0.245***	0.183***	0.206***	0.103***	1.000									
(6) AI	0.182***	0.128***	0.191***	0.047*	0.611***	1.000								
(7) Digital leader	0.094***	0.121***	0.092***	0.094***	0.404***	0.383***	1.000							
(8) CEO_President.	0.261***	0.150***	0.153***	0.089***	0.295***	0.245***	0.302***	1.000						
(9) CFO_COO	−0.062**	−0.004	−0.022	0.012	0.012	0.042	0.067	−1.000***	1.000					
(10) New CEO	0.189***	0.077***	0.133***	0.015	0.219***	0.266***	0.480***	0.356***	0.055	1.000				
(11) Aqcuisition	0.208***	0.068**	0.125***	0.009	0.214***	0.293***	0.514***	0.344***	0.021	0.774***	1.000			
(12) Sales dispersion	0.145***	0.083***	0.109***	0.038	0.141***	0.131***	0.211***	0.112***	0.089***	0.241***	0.264***	1.000		
(13) Total revenue	0.161***	−0.006	0.027	−0.036	0.163***	0.146***	0.015	0.176***	−0.083***	0.020	0.058**	0.098***	1.000	
(14) Employees	0.182***	−0.129***	−0.045	−0.141***	0.124***	0.098***	0.021	0.090***	0.008	0.045	0.079***	0.237***	0.464***	1.000
(15) Public firm	−0.022	−0.057**	−0.052*	−0.038	−0.058	−0.109*	−0.001	−0.159***	−0.106*	−0.278***	−0.061	−0.051*	0.059**	0.114***

Notes: ***, ** and * indicate significance at $p < 0.01$, 0.05 and 0.10, respectively (two-tailed).

[11] Small firms (by revenues) are screened from the sample, making it less likely that private firms are also systematically younger and more entrepreneurial than public firms.

[12] As a robustness check, we replace industry fixed effects with industry-level controls such as industry employee intensity, asset intensity, and return on assets (ROA) derived from Compustat and obtain similar results.

Table 6
The association of Technology implementation with Cost Reduction and Revenue Change.

Variables	(1)	(2)	(3)	(4)
	Cost reduction	Cost reduction	Revenue change	Revenue change
RPA	3.21***	2.57***	4.37***	2.76**
	(0.64)	(0.82)	(1.01)	(1.30)
AI	1.07*	0.45	1.33	−0.24
	(0.64)	(0.81)	(1.01)	(1.28)
RPA_AI_Joint		1.60		3.98**
		(1.28)		(2.02)
Digital leader	−0.30	−0.24	2.02*	2.17**
	(0.69)	(0.70)	(1.07)	(1.07)
CEO_President	3.46***	3.48***	4.48***	4.53***
	(0.66)	(0.66)	(1.03)	(1.03)
CFO_COO	−0.23	−0.26	1.35	1.28
	(0.66)	(0.66)	(1.03)	(1.03)
New CEO	0.20	0.25	0.36	0.46
	(0.88)	(0.88)	(1.38)	(1.38)
Acquisition	1.12	1.14	0.37	0.41
	(0.89)	(0.88)	(1.39)	(1.38)
New CEO * acquisition	1.60	1.55	−0.95	−1.07
	(1.25)	(1.25)	(1.95)	(1.95)
Sales dispersion	0.54	0.63	1.47	1.67
	(0.90)	(0.90)	(1.40)	(1.40)
Total revenue	0.20	0.20	0.20	0.20
	(0.19)	(0.19)	(0.29)	(0.29)
Employees	0.35***	0.34***	−0.87***	−0.87***
	(0.11)	(0.11)	(0.17)	(0.17)
Public firm	0.12	0.12	0.26	0.26
	(0.65)	(0.65)	(1.01)	(1.01)
Constant	−3.35	−3.36	10.99	10.99
	(5.33)	(5.33)	(8.15)	(8.14)
Industry FE	YES	YES	YES	YES
Country FE	YES	YES	YES	YES
R-squared	0.18	0.18	0.15	0.15
N	1193	1193	1219	1219

Notes: ***, ** and * indicate significance at p < 0.01, 0.05 and 0.10, respectively (two-tailed). Standard errors are in parentheses. Industry FE and Country FE denote industry and country fixed effects, respectively. RPA_AI_Joint = RPA × AI. All other variable definitions appear in Table 2.

Table 7
Decomposing Revenue change into Price change and Volume change.

Variables	(1)		(2)	
	Price change	**Volume change**	**Price change**	**Volume change**
RPA	1.74***	2.28***	0.91	1.59
	(0.43)	(0.80)	(0.55)	(1.03)
AI	1.27***	−0.02	0.47	−0.70
	(0.43)	(0.80)	(0.55)	(1.02)
RPA_AI_Joint			2.05**	1.70
			(0.86)	(1.61)
Digital leader	−0.02	1.86**	0.06	1.92**
	(0.46)	(0.85)	(0.46)	(0.85)
CEO_President	1.44**	2.70***	1.46***	2.72***
	(0.44)	(0.81)	(0.44)	(0.81)
CFO_COO	0.15	1.21	0.11	1.18
	(0.44)	(0.82)	(0.44)	(0.82)
New CEO	−0.16	0.52	−0.11	0.57
	(0.59)	(1.09)	(0.59)	(1.09)
Acquisition	−0.21	0.62	−0.18	0.64
	(0.59)	(1.10)	(0.59)	(1.10)
New CEO * acquisition	1.54*	−2.61*	1.48	−2.66*
	(0.83)	(1.55)	(0.83)	(1.55)
Sales dispersion	0.23	1.13	0.33	1.22
	(0.60)	(1.11)	(0.60)	(1.11)
Total revenue	−0.03	0.13	−0.03	0.13
	(0.12)	(0.23)	(0.12)	(0.23)
Employees	−0.19***	−0.64***	−0.20***	−0.64***
	(0.07)	(0.13)	(0.07)	(0.05)
Public firm	0.44	−0.03	0.44	−0.02
	(0.43)	(0.80)	(0.43)	(0.80)
Constant	5.55	7.67	5.55	7.67
	(3.47)	(6.47)	(3.46)	(6.46)
Industry FE	YES	YES	YES	YES
Country FE	YES	YES	YES	YES
R-squared	0.16	0.10	0.16	0.10
N	1218	1218	1218	1218

Notes: ***, ** and * indicate significance at p < 0.01, 0.05 and 0.10, respectively (two-tailed). Standard errors are in parentheses. Industry FE and Country FE denote industry and country fixed effects, respectively. RPA_AI_Joint = RPA × AI. All other variable definitions appear in Table 2.

country of origin (Table 1, Panel B).

Table 5 reports Pearson correlations between the firm-level variables (with tetrachoric correlations between dichotomous indicators). Many of the correlations are modest in size, indicating low risk of multicollinearity in the analysis (as is confirmed by low unreported VIF values of the subsequent regression analyses).

4.2. Results

Table 6, columns 1 and 2 show the regression results of estimating models (1) and (2) with *Cost reduction* as the performance outcome. Column 1 shows that consistent with prior research, the implementation of RPA and/or AI is positively associated with cost reduction ($p < 0.01$ and $p < 0.10$, respectively). However, column 2 shows that the interaction term, *RPA_AI_Joint,* is not significantly associated with cost reduction. Thus, while each technology is independently associated with cost reduction, their joint implementation is not associated with additional cost reduction. Accordingly, H1 is not supported.

Table 6, columns 3 and 4 show the results of estimating models (1) and (2) with *Revenue change* as the performance outcome. Column 3 shows that RPA implementation is associated with a significant positive change in revenue ($p < 0.01$), whereas the coefficient for AI is not statistically significant. Column 4 shows that joint implementation of RPA and AI, represented by *RPA_AI_Joint,* is positively associated with revenue growth ($p = 0.049$). This suggests that firms implementing both technologies experienced stronger revenue growth than firms implementing only one technology. Thus, H2 is supported.

Table 7 decomposes revenue change into its two underlying components: price change and volume change, which are regressed on the same variables as in Table 6. Column 1 of Table 7 reports the separate associations of RPA and AI, while column 2 extends the specification to include the interaction term, *RPA_AI_Joint*. In column 1, both RPA and AI are positively associated with price change (both $p < 0.01$), indicating that use of each technology independently is associated with higher output prices. In contrast, RPA has a positive association with volume change ($p < 0.01$), while AI is not associated with volume change ($p > 0.10$). When the joint implementation variable is introduced (column 2), the results show that the positive association between joint implementation of RPA and AI and revenue growth is primarily related to price increases. The coefficient on *RPA_AI_Joint* is positive and significant for price change ($p = 0.017$), while it is not statistically significant for volume change. This pattern suggests that the performance gains associated with joint implementation of RPA and AI operate primarily through price effects rather than increased sales volume.

Table 8 reports the results of estimating model (3) to test H3 and H4 by analyzing the association between firms' strategic intent in adopting the technologies and the corresponding performance outcomes. Consistent with the empirical approach described in Section 3, the analysis is conducted across three subsamples: firms that have implemented only RPA, only AI, and both technologies. Columns 1–3 present results for realized *Cost reduction.* For RPA adopters (column 1), implementing RPA with a stronger cost reduction intent is associated with greater realized cost reduction ($p < 0.10$). In contrast, for AI adopters (column 2) the association between cost-reduction intent and realized *Cost reduction* is not statistically significant. Among firms that have implemented both RPA and AI (column 3), we find no support for H3: strategic intent to reduce costs is not associated with additional cost

Table 8
Aligning technology implementation with the strategic intent of technology adoption.

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Cost reduction	**Cost reduction**	**Cost reduction**	**Revenue change**	**Revenue change**	**Revenue change**	**Price change**	**Volume change**
Sample Selection on implementation of:	*RPA*	*AI*	*RPA and AI*	*RPA*	*AI*	*RPA and AI*	*RPA and AI*	*RPA and AI*
Cost_RPA	1.70*		8.57***					
	(1.02)		(2.84)					
Cost_AI		0.78	−7.01**					
		(0.89)	(3.04)					
Cost_Joint			−0.84					
			(1.32)					
Rev_RPA				1.72		6.82	2.05	3.82
				(1.59)		(4.32)	(1.55)	(3.23)
Rev_AI					1.77	−7.67	−1.64	−5.51
					(1.78)	(5.33)	(1.91)	(3.98)
Rev_Joint						7.61**	1.42	5.92**
						(3.83)	(1.37)	(2.86)
Constant	−22.07	−24.74	−9.30	12.99	−33.77	−52.49	−4.46	−34.68**
	(14.56)	(16.29)	(21.24)	(24.55)	(27.52)	(39.03)	(14.01)	(29.15)
R-squared	0.33	0.28	0.49	0.24	0.19	0.27	0.33	0.25
Firm controls	YES	YES	YES	YES	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES	YES	YES	YES	YES
Country FE	YES	YES	YES	YES	YES	YES	YES	YES
N	289	296	156	292	301	159	159	159

Notes: ***, ** and * indicate significance at $p < 0.01$, 0.05 and 0.10, respectively (two-tailed). Standard errors are in parentheses. Cost_RPA, and Cost_AI in Column 3 and Rev_RPA, and Rev_AI in Columns 6 and 7 are mean centered measures. Thus, Cost_Joint in Column 3 is calculated as the product of the mean centered measures, Cost_RPA, and Cost_AI. Rev_Joint in Columns 6, 7 and 8 is calculated as the product of the mean centered measures, Rev_RPA, and Rev_AI. Industry FE and Country FE denote Industry Fixed Effects and Country Fixed Effects, respectively. All other variable definitions appear in Table 2.
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reduction when both technologies are used together ($p > 0.10$). Instead, the coefficient on *Cost_RPA* is positive and significant ($p < 0.01$), while the coefficient on *Cost_AI* is negative and significant ($p = 0.023$). This suggests that, among firms that have adopted both RPA and AI, a stronger emphasis on cost-reduction intent for using AI is associated with lower realized cost reduction. A plausible explanation is that applying AI alongside RPA to pursue cost reduction introduces additional coordination, technical, and governance complexity, which may offset potential cost reduction gains. These findings thus suggest that cost reduction is more effectively realized through RPA, whereas the addition of AI for the same intent may create frictions that limit gains.

Columns 4–6 of Table 8 present the results for revenue growth. Neither RPA adopters (column 4) nor AI adopters (column 5) show a significant association between a revenue growth intent and realized *Revenue change* (both $p > 0.10$). However, among firms that have implemented both RPA and AI (column 6), a stronger revenue growth intent is positively associated with realized revenue growth ($p = 0.049$). These findings provide support for H4 and indicate that firms that jointly implement RPA and AI with a clear focus on revenue growth are more likely to realize complementary revenue growth benefits.

Columns 7 and 8 further decompose revenue change into *Price change* and *Volume change*. Among firms jointly implementing RPA and AI, the positive association between revenue growth intent and realized revenue growth is primarily reflected in increases in volume ($p = 0.040$). This suggests that firms adopting RPA and AI with a clear revenue growth intent leverage the combined use of the technologies to expand activity volume levels, such as by alleviating operational bottlenecks and improving process scalability. It is important to note that these volume benefits apply to firms that purposefully implement both technologies for revenue growth and are additional to the price-related gains associated with joint implementation relative to firms not using both technologies, as found in Table 7.

Overall, the results of Study 1 indicate that while the independent use of RPA and AI is significantly associated with cost reduction, their joint implementation is not associated with additional cost reduction benefits, providing no support for H1. In contrast, combined implementation of RPA and AI is positively associated with revenue growth, primarily through price increases, supporting H2. Consistent with these findings, H3 is not supported as a higher emphasis on cost reduction intent is not associated with greater cost reduction gains from joint implementation. In contrast, H4 is supported: among firms using both RPA and AI, a stronger revenue growth intent is positively associated with greater revenue growth, primarily resulting from increases in activity volume. Taken together, these findings suggest that while RPA is effective for cost reduction, the primary value of combining RPA with AI arises when firms strategically use these technologies to support revenue growth.

5. Study 2

5.1. Methods

5.1.1. Research design and sample description

To provide further evidence on the association between joint implementation of RPA and AI and firm performance, we use a second survey dataset collected from large firms across 14 countries worldwide. As the survey questions differ partly from those of Study 1, they do not allow testing H3 and H4 and provide different measures for tests of H1 and H2. Thus, Study 2 can be considered to provide additional independent evidence on hypotheses H1 and H2. The survey evidence was collected by the survey firm from early June to early July 2020. Given that the survey was administered during the COVID−19 pandemic, questions explicitly addressed the period immediately before the pandemic as well as during it. To avoid influences of the pandemic on responses, we only use questions pertaining to the period immediately before the pandemic.

Survey respondents were 1089 senior executives from large firms in 14 countries. Similar to Study 1, the sampling criteria and data collection result in representativeness of the data and statistical results relating to large firms, across a wide range of industries and a range of countries. Table 9 reports descriptive statistics on how the sample firms are distributed over size categories, industries, and countries.

Table 9
Sample descriptive statistics Panel A of this table reports descriptive information about sales, employees, and industry distributions, and whether firms are private or publicly listed. Panel B shows the distribution of sample firms across the fourteen countries of location.

Panel A: Firm size, industry, and listing (N = 1089)					
Sales in millions $	*# obs*	*% obs*	*# Employees*	*# obs*	*% obs*
200 – 300	63	5.79	1–1000	68	6.24
300 – 500	66	6.06	1000–2500	207	19.01
500 – 750	71	6.52	2500 – 5000	194	17.81
750 – 1000	74	6.80	5000 – 10,000	219	20.11
1000 – 1500	122	11.20	10,000 – 25,000	170	15.61
1500 – 3000	121	11.11	25,000 – 50,000	106	9.73
3000–5000	136	12.49	50,000 – 100,000	71	6.52
5000–10,000	134	12.30	100,000 and more	54	4.96
10,000–20,000	125	11.48			
20,000–30,000	63	5.79			
30,000–60,000	45	4.13			
More than 60,000	69	6.34			
Principal industry					
Banking	67	6.15	Medical Technology	72	6.61
Capital Markets	41	3.76	Oil & Gas	73	6.70
Chemicals and Specialty Materials	60	5.51	Pharmaceuticals	57	5.23
Hardware and Semiconductors	55	5.05	Public Sector	53	4.87
Health Care Providers	60	5.51	Retail	81	7.44
Hospitality	52	4.78	Consumer products	67	6.15
Industrial products	83	7.62	Software and IT services	77	7.07
Insurance – General	41	3.76	Telecom	55	5.05
Insurance - Life and Annuity	26	2.39	Transportation	69	6.34
Publicly listed firm					
No	705	64.74			
Yes	384	35.26			

Panel B: Firm country of origin (N = 1089)					
Country	*# obs*	*% of sample*	*Country*	*# obs*	*% of sample*
Australia	44	4.04	Italy	70	6.43
Brazil	72	6.61	Japan	48	4.41
China - HK	44	4.04	Mexico	47	4.32
China - Main	78	7.16	New Zealand	10	0.92
France	51	4.68	Spain	70	6.43
Germany	79	7.25	United Kingdom	124	11.39
India	44	4.04	USA	308	28.28

5.1.2. Survey design and variable measurement

The survey built on the 2018 survey, offering insight into the clarity and understandability of recurring questions, and respondents' ability or knowledge to address questions. The survey, including newly added questions, was reviewed thoroughly by the academic authors and by the survey firm. Similar to the survey of Study 1, questions were presented in a fixed order, first probing the firm's current conditions, and then progressing to the future. Survey items within question blocks were presented in random order to guard against ordering effects. Again, the objective nature of most survey questions mitigates potential biases associated with perceptions (Golden, 1992), but also precludes providing statistical evidence on construct validity, which is primarily based on face validity. Below we describe the measurement of model variables. Table 10 provides a summary of variable measurement and descriptive statistics.

Expected cost reduction is measured using a survey item regarding the firm's annual cost reduction target as a percentage of costs immediately before the COVID−19 crisis (Table 11, Panel A). The annual cost reduction target question is captured with seven response categories, ranging from "less than 5%" to "75% or more". Given the scale construction, we use the lower bound of each interval to determine a specific cost reduction percentage (e.g., 1, 5, 10…75, see Table 11, Panel A), noting that using the interval midpoint instead yields similar results for the hypothesis tests.

Expected revenue growth is captured by the firm's 12-month business outlook for revenues as it was immediately before the COVID−19 crisis (Table 11, Panel B). The survey scale captures expected changes in revenue as percentage intervals. Similar to the treatment of *Revenue change* in Study 1, we use the lower absolute value of each interval with the sign reflecting the direction of change to approximate expected revenue change, and reverse-code the scale so that lowest (highest) percentage corresponds to anticipated decline (increase) in revenue. We exclude a small proportion of (30) firms reporting extreme expected growth of over 100% (see Table 11, Panel B), as these responses represent unusually extreme forward-looking growth expectations that are not comparable to the bounded interval categories used for the remainder of the scale. After this adjustment, the measure ranges from −76 to + 76 %. Most firms expected growth in revenue, although 5.26 % of sample firms expected a decline and 3.79 % expected no change.

Robotic Process Automation (*RPA)* is measured using a survey item capturing if RPA was among the top five most relevant technologies in the firm's operating model. Thus, *RPA* captures the firm's priority for RPA as one of the most important technologies among nine listed technologies: Cloud computing, Cognitive and artificial intelligence tools (AI), Cybersecurity solutions, Robotic Process Automation (RPA), Internet of things (IoT), Business Intelligence and advanced analytics, Virtual reality or remote enablement technology, Blockchain, and Other. *RPA* ranges from 1 to 6, with 1 indicating that RPA was not among the top five most relevant technologies in the firm's operating model (i.e., least important), and 6 indicating that RPA was the most important technology in the firm's operating model.

Artificial Intelligence (AI) is measured using a survey item that, similar to *RPA*, captured if AI was among the top five most relevant technologies in the firm's operating model, selected from the same list of nine technologies reported above. Like *RPA*, *AI* ranges from 1 (not among the top five most relevant technologies in firm's operating model) to 6 (most important in firm's operating model). As Table 10 shows, the mean score of RPA and AI among the sample firms is 2.62 and 3.71, respectively, indicating that these technologies are considered to be relatively important compared to other technologies.

Table 10
Variable measurement and summary statistics.

Variable	Description	Obs.	Mean	S.D.	Min	Max
Expected cost reduction	Cost reduction target (percentage) for the next 12 months	952	12.29	14.80	1	75
Expected revenue growth	Expected 12-month percentage change in revenue	1053	16.70	20.70	−76	76
RPA	Priority of RPA for the firm among nine listed technologies (from 1 = least important to 6 = most important for the firm's operational model).	1089	2.62	1.64	1	6
AI	Priority of AI for the firm among nine listed technologies (from 1 = least important to 6 = most important for the firm's operational model).	1089	3.71	1.95	1	6
CEO_President	Dummy variable is 1 when respondent is CEO or President of the company	1089	0.25	0.43	0	1
CFO_COO	Dummy variable is 1 when respondent is CFO or COO of the company	1089	0.25	0.43	0	1
Total revenue	Log of total revenues	1089	8.18	1.57	5.52	11.00
Employees	Log of employees	1089	0.74	1.38	−4.79	5.52
Public firm	Dummy variable is 1 for firms that are publicly traded, otherwise 0	1089	0.35	0.48	0	1
Government agency	Dummy variable is 1 for firms that are government agency, otherwise 0	1089	0.04	0.19	0	1
Private equity	Dummy variable is 1 for private organization funded by private equity or fund, otherwise 0	1089	0.04	0.19	0	1

We operationalize the joint implementation of RPA and AI based on the importance of both technologies in the firm's operating model. Specifically, we define *RPA_AI_Joint* as an indicator variable equal to 1 if both RPA and AI are ranked among the top four technology priorities and 0 otherwise.[13] This classification captures firms that have intentionally deployed both technologies as core elements of their operating model. Based on this definition, 34.7% of sample firms are classified as joint implementers of RPA and AI.

Table 11
Summary statistics for Expected Cost Reduction and Revenue Growth.

Panel A: Expected cost reduction based on cost reduction target within the next 12 months (%)

Cost reduction target (%)	***Scale value***	***# obs***	***% obs***
< 5%	1%	249	24.4
5–10%	5%	259	25.4
10–20%	10%	180	17.6
20–30%	20%	122	12.0
30–50%	30%	79	7.7
50–75%	50%	47	4.6
75% or more	75%	16	1.6
No specific target established	-	69	6.8

Panel B: 12-month expected revenue change

	Scale value	***# obs***	***% obs***		***Scale value***	***# obs***	***% obs***
Decline between 76% and 100%	−76%	1	0.1	Growth of greater than 100%	-	30	2.8
Decline between 51% and 75%	−51%	3	0.3	Growth between 76% and 100%	76%	48	4.4
Decline between 41% and 50%	−41%	2	0.2	Growth between 51% and 75%	51%	56	5.2
Decline between 31% and 40%	−31%	2	0.2	Growth between 41% and 50%	41%	72	6.7
Decline between 21% and 30%	−21%	10	0.9	Growth between 31% and 40%	31%	80	7.4
Decline between 11% and 20%	−11%	13	1.2	Growth between 21% and 30%	21%	149	13.8
Decline between 1% and 10%	−1%	26	2.4	Growth between 11% and 20%	11%	272	25.1
Anticipate flat top line	0%	41	3.8	Growth between 1% and 10%	1%	278	25.7

We control for several firm and industry characteristics (see Table 10) that may be related to the expected cost reduction and revenue growth associated with technology use. Most control variables are similar to those of Study 1. However, since the two survey instruments had some variation in questions and in the timing of data collection, there are some differences in the control variables.[14]

Similar to Study 1, *CEO_President* and *COO_CFO* indicate if the respondent is the firm's CEO or President (24.79 % of firms), or the COO or CFO (24.61 %), respectively. *Public firm* indicates if the firm is publicly traded (35.26 %). Since the survey includes more indicators of

[13] If AI and RPA are both indicated among the top four priorities for a firm's operating model (i.e., above a 'low importance'), then it is likely that the firm relies at least to some extent on these technologies. We also set this threshold of being among the top four priorities to ensure that both technologies are among the firm's topmost relevant technologies out of all nine types potentially used.

[14] For instance, the instrument of Study 2 did not capture the presence of a digital leader, CEO change or acquisitions, but did provide more refined measures of firm governance.

Table 12
Correlation table of firm-level variables This table reports the Pearson correlation between pairs of firm-level variables. Cells of paired indicator variables (Variables 5, 6, 9, 10, and 11) contain tetrachoric correlations.

	1	2	3	4	5	6	7	8	9	10
(1) Expected cost reduction	1.000									
(2) Expected revenue growth	0.148***	1.000								
(3) RPA	0.112***	0.073**	1.000							
(4) AI	0.169***	0.163***	0.058*	1.000						
(5) CEO_President	0.183***	0.147***	0.126***	0.014***	1.000					
(6) CFO_COO	0.036	0.081***	0.006	0.026	−1.000***	1.000				
(7) Total revenue	0.045	−0.034	−0.058*	−0.005	0.017	−0.097***	1.000			
(8) Employees	0.009	0.020	0.051*	0.078**	−0.030	−0.035	−0.550***	1.000		
(9) Public firm	−0.149***	−0.045	−0.031	0.023	−0.202***	−0.091	0.193***	0.066**	1.000	
(10) Gov. agency	−0.041*	−0.005	−0.100***	−0.068**	−0.383***	−0.147	0.018	−0.027	−1.000***	1.000
(11) Private equity	0.117***	0.045	0.011	−0.067**	0.076	0.014	−0.041	−0.006	−1.000***	−1.000***

Notes: ***, ** and * indicate significance at p < 0.01, 0.05 and 0.10, respectively (two-tailed).

Table 13
The association of Technology use with Expected Cost Reduction and Expected Revenue Growth.

Variables	(1)	(2)	(3)	(4)
	Expected cost reduction	Expected cost reduction	Expected revenue growth	Expected revenue growth
RPA	0.88***	0.81**	0.33	−0.20
	(0.29)	(0.37)	(0.38)	(0.50)
AI	0.94***	0.87***	0.96***	0.51
	(0.24)	(0.32)	(0.32)	(0.42)
RPA_AI_Joint		0.51		3.45*
		(1.54)		(2.05)
CEO_President	5.56***	5.56***	6.24***	6.26***
	(1.22)	(1.22)	(1.65)	(1.64)
CFO_COO	3.56***	3.57***	5.82***	5.90***
	(1.16)	(1.16)	(1.54)	(1.53)
Total revenue	1.19***	1.19***	−0.38	−0.36
	(0.38)	(0.38)	(0.51)	(0.51)
Employees	0.70*	0.70*	0.04	0.04
	(0.42)	(0.42)	(0.55)	(0.55)
Public firm	−4.52***	−4.53***	−1.08	−1.06
	(1.10)	(1.10)	(1.46)	(1.46)
Government agency	−3.83	−3.84	−6.29	−6.31
	(3.27)	(3.28)	(4.10)	(4.10)
Private equity	6.28***	6.29***	1.11	1.14
	(2.33)	(2.33)	(3.23)	(3.23)
Constant	−8.46*	−8.25*	19.47***	20.88***
	(4.35)	(4.40)	(5.70)	(5.75)
Industry FE	YES	YES	YES	YES
Country FE	YES	YES	YES	YES
R-squared	0.14	0.14	0.15	0.15
N	952	952	1053	1053

Notes: ***, ** and * indicate significance at p < 0.01, 0.05 and 0.10, respectively (two-tailed). Standard errors are in parentheses. Industry FE and Country FE denote industry and country fixed effects, respectively.

governance, we additionally control for whether the organization is a *Government agency* (3.58 %) or is a private organization funded by *Private equity* (3.94 %). We again control for firm size, based on the log of annual revenue (converted to Euro amounts) in the last fiscal year (*Total revenue*), and the log of number of employees worldwide (*Employees*).[15] To control for unobserved, time-invariant heterogeneity at the industry level, we also include industry fixed effects.

Table 12 reports Pearson correlations between the firm-level variables. The modest size of most correlations indicates low risk of multicollinearity in the analysis (as is confirmed by low unreported VIF values of the subsequent regression analyses).

5.2. Results

Table 13, columns 1 and 2 present the results of regression models (1) and (2) from Section 3, with *Expected cost reduction* as the performance outcome. Column 1 indicates that both *RPA* and *AI* are individually positively associated with expected cost reduction (both $p < 0.01$). Column 2 shows that their combined implementation, indicated by *RPA_AI_Joint,* is not statistically significant. Thus, while each technology is independently associated with expected cost reduction, their joint implementation is not. Accordingly, H1 is not supported. These findings converge with the results on realized cost reduction in Study 1 (see Section 4.2).

Columns 3 and 4 of Table 13 show the results of regression models with *Expected revenue growth* as the performance outcome. As column 3 shows, *AI* is positively associated with *Expected revenue growth* ($p < 0.01$), whereas *RPA* is not statistically significant. Column 4 indicates that when *RPA_AI_Joint* is included, the coefficient for AI is no longer statistically significant, while the coefficient for *RPA_AI_Joint* is positive and marginally statistically significant ($p < 0.10$). This provides directionally consistent although somewhat weaker evidence for H2, suggesting that expected revenue growth is more strongly associated with the joint implementation of RPA and AI rather than with the independent implementation of AI alone. Firms that implement both technologies, thus, report higher expected revenue growth than firms that implement only one of the technologies. These findings are again consistent with the realized revenue growth results found in Study 1 (see Section 4.2).

6. Discussion

This study provides large-scale empirical evidence on the association between the combined implementation of Robotic Process Automation (RPA) and Artificial Intelligence (AI) and firm financial performance. Using two international datasets, we examine whether firms that implement both technologies experience greater performance benefits than firms implementing only one technology. We find a consistent pattern across both studies. In Study 1, RPA and AI are individually associated with cost reduction over a one-year period, but their combined implementation is not significantly associated with additional reduction in cost. In contrast, combined implementation is positively related to additional revenue growth over a 24-month period, with this association linked primarily to price increases. Among firms that jointly implement RPA and AI with a stronger revenue growth intent, higher revenue growth is also reflected in volume increases. We observe similar patterns in Study 2 for expected cost reduction and expected revenue growth in the upcoming year.

Our findings suggest that associations consistent with theorized complementarities between RPA and AI emerge primarily for revenue

[15] Again, respondents selected intervals for these measures (e.g., 4 = 5000 to less than 10,000 employees; 5 = €270 million to less than €450 million annual revenue). Similar to Study 1, we transform these intervals to specific values using the low-point, noting that estimation results are similar when using the mid-point of the scale (with the low point for the final open-ended category).

growth rather than for cost reduction outcomes. Although both technologies are independently associated with cost reduction, our findings indicate that their combined implementation is not associated with additional cost reduction beyond these independent associations. One possible explanation is that RPA and AI optimize different, non-overlapping dimensions of input use: RPA primarily reduces labor and error-related costs by automating standardized processes, while AI enhances decision quality, prediction accuracy, and customer interactions. Moreover, integrating these technologies may require substantial coordination, process redesign, and organizational change, which may offset potential efficiency-related benefits. The organizational and technical complexity of integration, together with high implementation and maintenance costs, may therefore explain why we do not observe additional cost reduction associated with combined implementation. Thus, while the results are consistent with revenue-enhancing complementarities, they do not provide evidence of comparable complementarities for cost reduction.

Our analysis of strategic intent provides additional insight into how firms' motives for adopting RPA and AI are associated with performance outcomes. We find no evidence that aligning RPA and AI around cost-reduction purposes is associated with greater cost reduction. Instead, among firms that have implemented both technologies, greater emphasis on using AI for a cost reduction intent is even associated with lower realized cost reduction. One possible explanation is that cost-focused AI initiatives may introduce additional coordination and implementation complexity that offset incremental cost reduction beyond what RPA already achieves for standardized tasks. In contrast, firms with a stronger revenue growth intent for implementing both RPA and AI exhibit higher revenue growth. In the overall sample, joint implementation is primarily associated with price increases relative to firms implementing only one technology. Among joint implementors with a stronger revenue growth intent, additional revenue growth is reflected by higher activity volume. Taken together, these findings suggest that while joint implementation is generally associated with price-related revenue growth, firms with a revenue growth intent may additionally gain revenue growth through expanded activity volume. More broadly, the results suggest that the alignment between strategic intent and the firm's targeted performance outcomes is an important contextual factor shaping the performance implications of technology implementation.

This study makes several theoretical and practical contributions. First, our findings extend prior qualitative research on the complementarities of digital technologies (Chakraborti et al., 2020; Anagnoste, 2018; Reddy et al., 2019) by providing large-scale quantitative evidence on how the independent and combined implementation of RPA and AI are associated with different dimensions of firm performance. Our evidence suggests that while RPA and AI are individually beneficial, their combined association with revenue growth is not just additive but consistent with complementarity. The results of analyses of two large international survey datasets suggest that the combined implementation of RPA and AI is associated more strongly with revenue enhancement rather than cost reduction. This finding qualifies the broader claims in practitioner and academic discussions that intelligent automation technologies necessarily generate broad productivity gains across all dimensions of performance.

Second, our revenue change decomposition analyses suggest that the positive association between joint implementation and revenue growth is linked primarily to price increases and, among firms with a stronger revenue growth strategic intent, also to volume increases. While we cannot directly identify the mechanisms underlying these associations, the findings are consistent with the idea that firms implementing both technologies may use them in ways that support differentiated offerings, customer responsiveness, or enhanced service capabilities. At the same time, these interpretations remain suggestive and should be treated cautiously because the underlying mechanisms are not directly observed in the data.

Third, our study also contributes to research on the alignment of technology use with organizational purpose (Lee et al., 2022), demonstrating that firms emphasizing a revenue growth intent for implementing RPA and AI exhibit greater revenue growth outcomes from combined implementation, whereas a stronger cost-reduction intent is not associated with greater cost reduction. For practitioners, these findings suggest that the organizational strategic intent motivating combined RPA and AI implementation may matter for the types of performance outcomes firms subsequently observe. More broadly, the results are consistent with the benefits of a strategic, goal-oriented approach to automation technology adoption and implementation. At the same time, strategic intent alone may not be sufficient. Organizational factors, such as a supportive digital culture, may play an important role in translating strategic intent into performance gains. In particular, a supportive digital culture fosters experimentation, learning, and employee engagement, which prior research suggests could enable unlocking the aspired benefits of intelligent automation (Wang & Zhang, 2025b). Future research should therefore examine how strategic intent and digital culture jointly relate to the performance outcomes associated with combined RPA and AI implementation.

In practitioner discourse, the combined use of RPA and AI is often described as "intelligent process automation", a term typically associated with truly integrated automation architectures that combine RPA's rule-based execution with AI's cognitive decision-making. A key limitation of our study is that the surveys capture firm-level combined implementation rather than confirmed process-level integration. This approach is consistent with prior research on IT complementarities, which similarly infers synergy potential from firm-level co-adoption (e.g., Bresnahan et al., 2002; Brynjolfsson & Hitt, 2000; Aral et al., 2012), but it does not allow us to observe how, where, when, or to what extent the two technologies interact. RPA and AI could, for example, be implemented independently, e.g., in different units, at different stages, or at different times, thereby limiting or delaying their complementarities.

Accordingly, our findings should not be interpreted as direct evidence that firms have achieved fully integrated intelligent automation architectures. Rather, the findings indicate that firms reporting implementation of both technologies also report stronger revenue-related outcomes, particularly when they emphasize a revenue growth intent. These patterns are consistent with the idea of firms making deliberate efforts to integrate the technologies intelligently in ways supportive of revenue growth, although the data do not allow direct observation of the underlying integration processes. At the same time, the absence of additional cost reduction associated with combined implementation suggests that merely implementing both technologies may not be sufficient to observe stronger cost-related outcomes. Instead, cost reduction complementarities may require deeper process integration, underscoring the importance of distinguishing between combined implementation and fully integrated automation systems.

Importantly, the data used in this study were originally collected by a practitioner organization for managerial insight and benchmarking rather than for academic research. As a result, survey design and measurement were not tailored to the specific theoretical objectives of studying complementarities of RPA and AI, which limits construct precision and coverage. In particular, AI is captured at an aggregate level and does not distinguish among different AI technologies or use cases. Future research could build on our findings using a purpose-built academic survey with more detailed measures of RPA and AI technologies and their use along with clearer distinctions between combined implementation and actual process-level integration. Such purpose-built data collection would also improve reproducibility by increasing transparency and control over measurement design.

We also caution that the assessed performance outcomes reflect different time horizons. Study 1 captures realized cost reduction and revenue change over periods of one and two years, respectively, limiting direct comparability of temporal effects. In Study 2, both expected cost reduction and expected revenue change are measured consistently over the same one-year period, but the data does not capture longer-term expected outcomes. Future research could capture multiple time-frames to better assess the short- and longer-term associations between technology implementation and firm performance.

Another limitation is that the study treats firms as isolated entities, whereas business processes are embedded in broader value chains and ecosystems. The performance implications of technologies may, therefore, depend on inter-organizational coordination with suppliers, partners, and customers. Prior research indeed shows that inter-organizational cooperation can amplify the impact of digital initiatives on innovation outcomes (Wang & Zhang, 2024), suggesting an important avenue for future research on automation complementarities.

The cross-sectional character of our datasets is another limitation that restricts causal inference and raises the possibility of reverse causality, whereby high-performing firms may be more likely to invest in both technologies. Accordingly, the study should be interpreted as documenting associations rather than causal effects. Despite this limitation, similar research designs are commonly used in studies examining broad firm-level relationships between technology and firm performance (for example, see Battisti & Stoneman, 2005; and more recently, Cho et al., 2023, and Ameye et al., 2023). Moreover, the use of two surveys conducted at different points in time (2018 and 2020), enables us to examine whether the observed patterns are consistent across independent samples, and helps mitigate this limitation. Future research could employ longitudinal or panel datasets to draw causal inferences about the impact of joint implementation of RPA and AI on firm performance outcomes.

Moreover, the datasets focus on large firms, which means that our findings may not generalize to small and medium-sized enterprises (SMEs; Wang & Zhang, 2025b). SMEs face very different resource constraints, technical issues, and implementation challenges when combining RPA and AI. The cost and organizational complexity of implementing and integrating both RPA and AI may be prohibitive for many SMEs, or the performance outcomes could differ substantially. Future research should examine whether similar performance complementarities emerge in SME contexts.

A limitation arising from our study design choices is that while our analyses provide insight into how process automation technologies relate to different dimensions of firm performance, they do not offer evidence on downsides, risks and ethical dilemmas associated with such technologies. For example, AI-driven systems can create "ethical anxiety" among employees and managers regarding job security, data privacy, and algorithmic fairness (Wang & Zhang, 2025a). This anxiety can act as a powerful barrier to effective implementation and moderate the association between technology implementation and performance. In line with emerging studies on the 'dark side' of AI (e.g., Ashok et al., 2022; Hai et al., 2025; Nunes et al., 2025; Wang & Zhang, 2025a), future studies of intelligent automation technology could incorporate such effects to provide a more balanced view on its benefits and costs.

Finally, our study covers firms' use of business process automation technologies before the recent emergence of Generative AI (GenAI) technologies, such as advanced large language models like ChatGPT. We therefore position this study as a baseline analysis of RPA–AI complementarities in the pre-Generative-AI era. Rather than being outdated, our findings provide an essential reference point for understanding how automation complementarities evolve as GenAI becomes embedded in organizational processes. Future research should examine whether GenAI changes the nature of these associations by reducing integration costs, altering implementation costs, or strengthening revenue-related and efficiency-related outcomes associated with combined technology implementation.

CRediT authorship contribution statement

Henri Dekker: Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **ANDERSON Shannon:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Arshad Farah:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Omar Aguilar:** Resources, Methodology, Investigation.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors did not make use of generative AI or AI-assisted technologies.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ijinfomgt.2026.103097.

Data availability

The authors do not have permission to share data.

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